Consequa-X: An Evidence and Intervention Protocol for Consequential Agent Actions

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***Abstract*—A lost acknowledgement can turn a successful agent action into a duplicate when retried. Consequa-X separates reported assessment, verified state and source-supported effect origin, preserving observation-specific provenance and explicit unknown verdicts. Its research adapter declares an on-reach intervention game and refusal-aware accounting around existing contrastive and permutation Shapley estimators. Across 2,048 scripted-simulator sampling trials, all 655,360 planned branches were retained. At one retry coefficient, normal intervals covered the reference in 118/128 trials; Hoeffding bounds covered in 128/128 but detected its sign in 8/128. A three-model factual study completed 503/504 episodes using 896 local calls. An equally informed, independently implemented reference agreed throughout; state verification made 431 correct definite claims, compared with 53 errors among 201 response-only claims. A same-evidence local judge made 9 errors among 60 definite state claims across 84 traces. Model-free transfer retained 656 retail/airline task-control cells and 900 operation captures, separating native whole-task DB equality from immediate field contracts. A pinned native Causal Agent Replay comparison retained 1,024 paired groups: all 640 eligible point vectors agreed exactly; 384 pairs remained unavailable. A collector deviation limits refused-attempt accounting in 64 already ineligible groups. These experiments use Consequa-X 1.1.0. Separately frozen offline migration to 2.0 preserved canonical conclusions on all 503 retained traces, including 72 unknown states, while diagnosis codes changed on 275. These conditional results support an inspectable protocol, without estimator novelty or general superiority; real-service effectiveness and independent human validation remain open.**

***Index Terms*—agent systems, outcome verification, evidence provenance, counterfactual replay, software artifacts, uncertainty**

I. Introduction

A consequential tool operation has distinct stages: selection, dispatch, execution, response, reported assessment and external observation. A service can commit an effect before its acknowledgement is lost. Retrying can create another effect unless the logical operation's identity and idempotency semantics are retained. Conversely, a successful response can coexist with an incomplete or incorrect effect. Explaining such runs requires more than reading the last tool response or the agent's prose.

State satisfaction and effect origin also differ. An external refund may satisfy the requested balance or count while remaining unlinked to this agent. A source-supported origin link is a provenance claim; it does not establish counterfactual necessity. A replay effect is a separate claim about a declared intervention, continuation policy and outcome distribution.

Consequa-X makes these distinctions inspectable. Its core contains typed records, schema validation, application verification and evidence-only explanation. Its research adapters provide isolated refund worlds, public-only policy inputs, a local Ollama adapter and bounded replay experiments. Reported assessments are captured outputs, not measurements of hidden reasoning.

The paper centers three testable protocol claims:

1) **C1: State and origin remain separate.** A satisfied contract can retain an other-actor or unknown origin. Successive readbacks retain their own supporting witnesses rather than transferring provenance between observations. Verification and graph tests exercise external-actor, missing-provenance and conflicting-witness cases; the local validation contains an other-actor example.
2) **C2: The replay game is explicit.** On-reach semantics define how normal early termination treats later requested holds. Strict and on-reach results remain separate. Analytic enumeration and the variable-length controls test this distinction; Appendix A states its formal scope.
3) **C3: Refusal remains observable.** Every planned branch is retained with call accounting; incomplete evidence or an invalid draw refuses affected estimates. The repeated-sampling study accounts for all 655,360 slots and reports availability beside numerical accuracy.

These are implementation and declared-game claims. The contribution combines them in a portable protocol around existing estimators. The matched factual study below finds exact agreement with an equally informed independent reference, supporting implementation correctness without empirical advantage. Figure 1 separates the captured-evidence core from the explicitly invoked research replay path.

II. Related Work

Causal Agent Replay (CAR) performs same-policy interventions on recorded decisions, evaluates user-specified outcomes and uses contrastive suffix and permutation Shapley attribution [1]. Consequa-X implements an independent binary adaptation with its own typed-world, evidence and refusal boundaries. Applying a verified outcome to replay is therefore not an estimator novelty claim. Policy-Conditioned Counterfactual Credit studies changes in final verified success under specified interventions and a continuation policy [2]; its learning objective differs from post-hoc external-effect diagnosis, but verified targets and valid interventions are shared concerns.

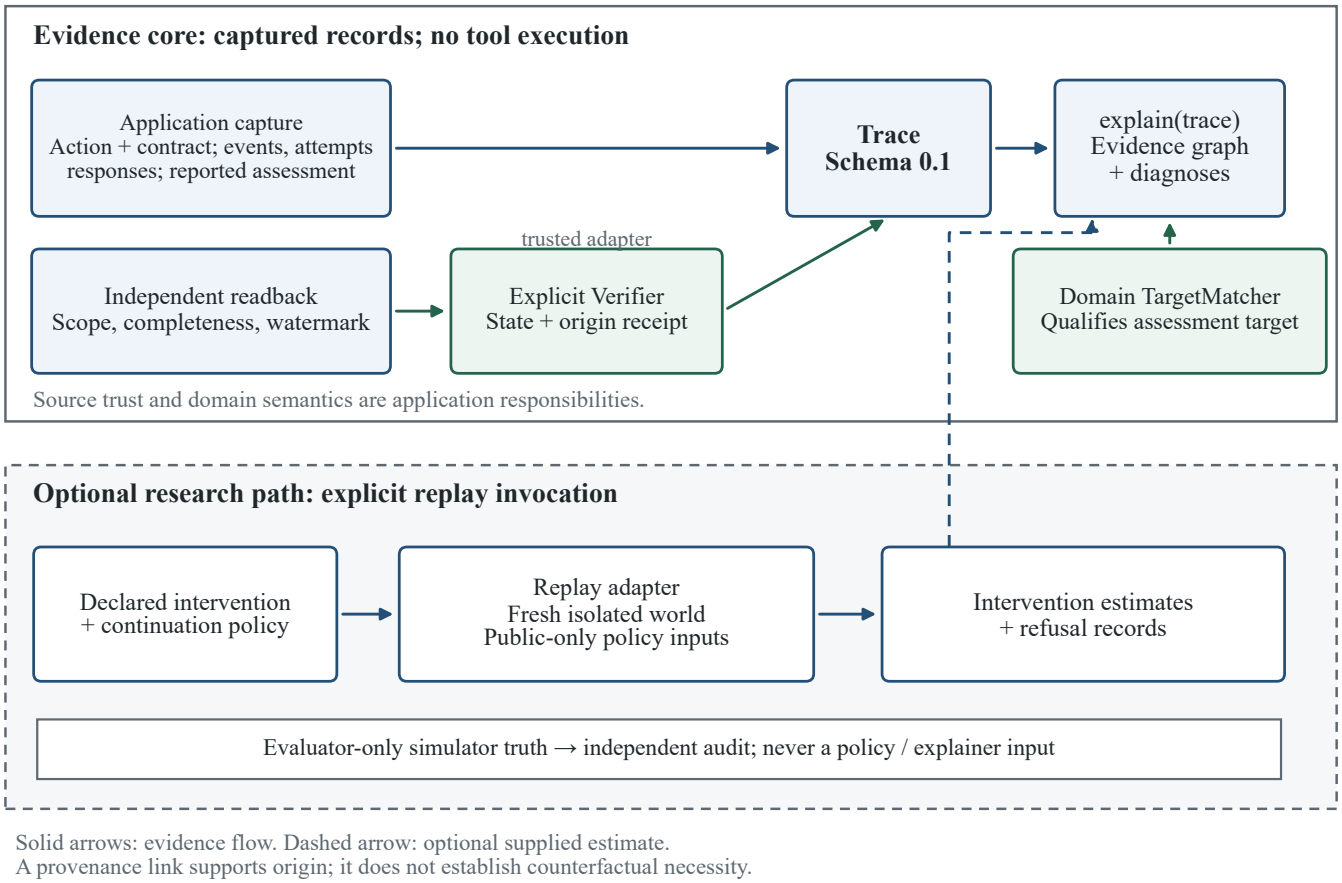


Fig. 1. Evaluated 1.x evidence flow and trust boundaries. Application verifiers interpret independent readback; explanations use captured evidence. Explicitly invoked replay can supply estimates, while evaluator truth remains audit-only. Source trust and domain semantics remain application responsibilities.

CausalLoss-Fin jointly intervenes on agent decisions and repairable infrastructure messages [3]. Its deterministic programmatic evaluation limits transfer to stochastic language policies, but it already occupies the broad combination of financial effects and joint decision/fault attribution. A future comparison must expose applicable variables to every method rather than charge omitted infrastructure effects to decisions.

Verified Tool Calls distinguishes acknowledgements from independently checked effects and uses unknown postcondition conclusions [4]. Callability Is Not Operability studies controlled interface ambiguity and terminal agent claims [5]. Three-valued verification and response/effect separation are foundations and comparison requirements here. These targeted primary-source checks do not constitute a systematic review or an independent reproduction of those papers' empirical results.

Who&When attributes failed multi-agent runs to an agent and decisive step from annotated logs [6]; this differs from verifying a particular external effect from an independent witness. PROV-DM already represents entities, activities, agents and responsibility [7]. Consequa-X is neither a new general provenance model nor a PROV conformance implementation. Its narrower contribution is an executable protocol linking temporal contracts, observation witnesses, separate origin verdicts and explicit refusal.

Agent–tool anomaly work already separates external events from runtime observations, including exogenous effects [8]; LIMBO grades committed-effect histories and analyzes late commits [9]. Response/effect separation and effect ledgers are therefore prior concepts. Our study audits a particular protocol rather than establishing a safer recovery controller. LIMBO's late-commit argument assumes a keyless, non-idempotent write with no known finite in-flight bound and finite task-completion time. Under those conditions, bounded watermark polling cannot guarantee exactly-once execution. Idempotence and eventual consistency are established systems concerns [10], [11].

AppWorld, ToolSandbox and τ-bench already evaluate world-state outcomes or stateful trajectories [12], [13], [14]. The maintained tau3 task corpus supplies the retail/airline tool-state transfer experiment below [20]; AppWorld, ToolSandbox and autonomous benchmark evaluation remain future comparisons. For statistics, known binomial Wald-interval limitations [15] motivate careful interval checks but do not by themselves establish paired-Shapley undercoverage. Modern

bounded-mean betting intervals [16] are planned comparators, with the sampling unit and stopping rule still requiring explicit validation for this replay setting.

III. Evidence Protocol and Core Interfaces

A. Objects, contracts and trust

The empirical studies below evaluate the versioned 1.x reference implementation and schema 0.1, with study-specific source pins. The current 2.x line uses schema 1.0, multiple actions per trace and explicit time bases. It refuses direct loading of 0.1 traces and provides an explicit, conservative migration path. Historical interfaces and results must therefore remain distinct from current product capabilities [23], [24].

Let H be visible history, A a logical action, E an execution attempt, R its response, B a timestamped reported assessment and O an independent readback observation. A contract C declares payload, count, deadline and stability horizon. The verifier conclusion $V(C, O)$ is satisfied, violated or unknown. Origin is separately confirmed, other actor or unknown. Evaluator-only simulator truth never becomes a policy or explainer input.

The built-in refund predicate uses integer currency minor units. An eligible effect matches payment, amount and currency, commits in the declared time window and persists through the horizon. Exact-count contracts also constrain the count during the stability interval. A stale watermark or inadequate scope and completeness yields unknown rather than a confident absence claim. Source authority, scope, completeness and watermark declarations remain trusted adapter inputs. Schemas and hashes validate structure and changed bytes, not source authenticity or real transactional guarantees. Figure 2 illustrates why a response and a reported assessment cannot substitute for a horizon-covering outcome observation.

B. Reusable application boundary

The evidence core can operate on a production application's captured records without any replay. A read-only adapter obtains independent observations; verification returns state and origin conclusions, and explanation renders the supporting evidence. This path requires neither simulator reset nor a second execution of a consequential action.

Applications can supply an explicit read-only verifier through `verify(action, observation, verifier=...)`. The wrapper checks receipt type, contract identity, observation identity, source, watermark and timing. Domain predicate semantics, authentication and read-only adapter behavior are the application's responsibility. There is no evaluation of arbitrary predicate strings. An offline artifact-delivery example constructs captured records and checks independently readback content digests without importing the refund simulator or executing writes.

In the evaluated 1.x interface, `explain(trace)` reads a captured verification and performs no replay. Refund target matching is built in; other predicates receive a neutral explanation unless a trusted domain matcher explicitly qualifies attributable effects against payload and temporal rules. Matchers can return only unique IDs within the receipt's attributable IDs. Unknown receipts are not interpreted as target matches. A qualified provenance link still is not a causal effect. In 2.0, verifiers record target-effect IDs and findings in each receipt; explanation consumes those records per action rather than invoking a separate target matcher. This API change requires explicit migration and version-specific conformance checking.

C. Successive witnesses and graph meanings

Repeated readbacks can disagree about the same effect identifier. The graph therefore retains observation-specific witnesses and an identity-only shared effect node. Dispatch-to-effect edges attach to the witness supplying origin metadata; verification edges use witnesses in that receipt's own observations. A later conflicting actor, corrected payload or reversal cannot overwrite an earlier witness and inherit its unqualified origin link. Shared identifiers do not authenticate or reconcile the records. Estimated-influence edges require captured event identifiers and an explicitly supplied intervention estimate.

IV. Replay Targets, Games and Uncertainty

A. Targets and conditional suffix effects

The refund adapter exposes three targets: initial action selection, global state-contract satisfaction and action-linked consequence. A known global violation is false for the linked target. Satisfied state with exclusively external qualifying effects is false for that target; insufficient provenance can instead yield unknown. The linked target combines global satisfaction and qualifying operation-linked witnesses rather than reconstructing an agent-only world. Unknown observations do not acquire numeric labels.

For known binary target Z, set bad outcome $X = 1 - Z$. Prefix replay holds factual decisions before position k and resamples the suffix. With one observed factual bad label x_f, the conditional estimand is

$$\gamma_k = \mathbb{E}[X \mid \text{prefix held, suffix resampled at } k] - x_f. \quad (1)$$

The factual term is one observed constant, not a population rate estimate. This factual-anchored target answers: how often would a fresh suffix avoid the bad outcome observed in this episode? It is useful for episode diagnosis, but does not estimate the effect of changing one decision against an independently resampled original arm. That population contrast would instead compare two expectations, for example $\mathbb{E}[X \mid \text{hold } k] - \mathbb{E}[X \mid \text{resample } k]$, with the other conditions declared. Neither estimand is generally preferable; they answer different questions and have different sampling costs. Here negative γ_k indicates rescue relative to the observed episode. An earlier irrelevant position can have a nonzero suffix effect because its suffix contains a later stochastic choice. The *pivot* is the latest position with negative population suffix effect under this definition. The retained nominal selector attempts to detect that position only if all planned positions are estimable. Its pointwise intervals give no simultaneous or post-selection

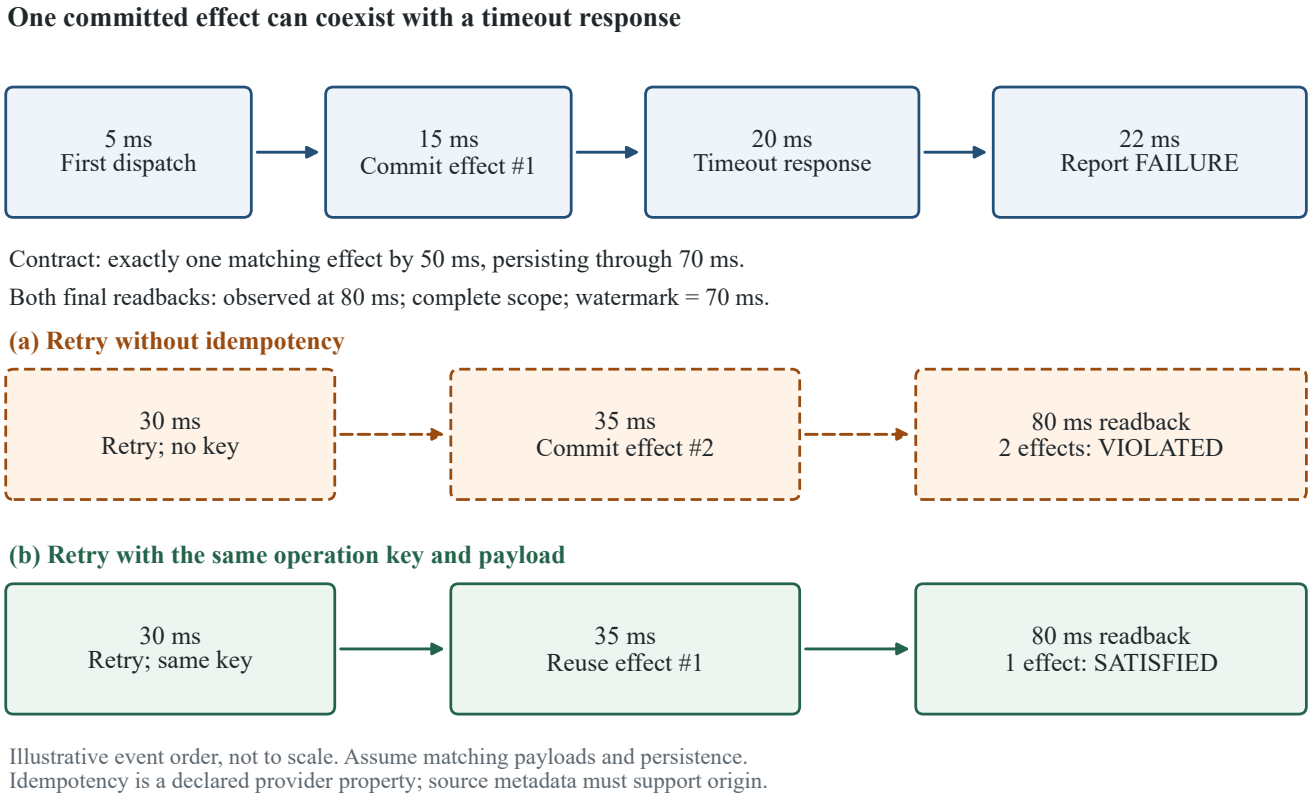


Fig. 2. Illustrative timeout-after-commit sequence. A committed effect precedes a timeout and reported failure. An unsafe retry creates a duplicate; reusing the same provider-scoped idempotency key and payload preserves one effect. Complete final readback covers the stability horizon. Arrows show event order, and times are illustrative rather than sampled observations.

coverage guarantee; selected pivots are exploratory diagnostics rather than unique causes.

B. Held-decision coalition games

Let N be reached factual positions. Coalition S requests holding captured decisions at positions in S; other decisions are sampled. Reached held decisions must match the typed public stage. In the *strict* game, an earlier termination leaving a requested held position unreachable invalidates the draw. Finite samples avoiding invalidity do not define the population game.

The separate *on-reach* game treats requested positions beyond a normal termination as inactive and uses the actual completed episode. Reached incompatible operations remain invalid; provider failures and exhausted budgets are not normal inactivity. This is an explicit alternative game, not a repair of retained strict results.

For a defined game $v(S) = \mathbb{E}[X \mid S]$, signed Shapley values allocate the held-decision effect [17]:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (2)$$

Fresh forward/reverse permutation walks form one paired contribution vector. Independent empty/full draws diagnose the endpoint scale. Values are signed allocations under the declared game, not normalized responsibility shares. Positive values mean holding the factual decision increases bad outcomes; negative values mean it is protective.

C. Intervals and refusals

Contrastive defaults use exact binomial Clopper–Pearson bounds translated by $-x_f$ [18]. Signed pair observations lie in $[-1, 1]$; the default Hoeffding half-width for P independent pairs and failure allowance α is [19]

$$h = \sqrt{\frac{2\log(2/\alpha)}{P}}. \quad (3)$$

This is the half-width before clipping the interval to $[-1, 1]$. Endpoint differences use two exact binomial marginals with a split failure allowance. Each marginal has confidence $1 - \alpha/2$; the difference interval is $[L_f - U_e, U_f - L_e]$. The union bound requires valid marginal bounds, not independence between the two arms. Bootstrap and normal intervals remain explicit legacy comparators. These are standard conditional methods with sampling assumptions and precision costs, not distribution-free guarantees for dependent model continuations. All reported coefficient intervals are pointwise. For a predeclared family of m valid marginal intervals, allocating failure probability α/m to each would give a Bonferroni family guarantee without independence across positions. That adjustment was not used in these retained studies. A nominal

latest-rescue selection therefore does not inherit a 95% family guarantee. Variance-adaptive or betting-based bounded-mean methods [16] are relevant future comparators; the present results do not establish that Hoeffding is optimal or that conservative uncertainty must have this width.

Replay requires a resettable, isolated simulator or a validated digital twin. It must not reissue real refunds or other consequential production writes. The studies below reset a scripted payment simulator, whose fidelity is a separate assumption; the evidence-only production path does not require it. Held replies consume no fresh calls; every attempted fresh call, including a provider/protocol failure, consumes budget. Every planned branch remains recorded. Unknown, invalid, failed and unstarted branches are not dropped or redrawn; affected estimates are refused. Targets and interval variants reuse captured evidence without extra policy calls.

D. Analytic refund reference

Consider the declared on-reach refund mechanism with adequate budget, complete horizon-covering readback and, for the linked target, sufficient provenance. The planted factual decisions are selection s, retry r and final stop t. Fresh selection occurs with probability p and fresh timeout retry with probability q under the declared independent choice law. The first attempt commits despite its timeout; without idempotency, a retry commits a second refund. Exactly one effect therefore occurs with probability $p(1-q)$. For the consequence bad-outcome game, the four relevant coalition values are

$$\begin{aligned} v(\varnothing) &= 1-p(1-q), & v(\{s\}) &= q, \\ v(\{r\}) &= 1, & v(\{s,r\}) &= 1. \end{aligned} \tag{4}$$

The final stop is a dummy: adding t changes none of these values. Averaging the two orders of the active players gives

$$\begin{aligned} \phi_s &= -\tfrac{1}{2}(1-p)(1-q), \\ \phi_r &= \tfrac{1}{2}(1+p)(1-q), \qquad \phi_t = 0. \end{aligned} \tag{5}$$

Their sum is the endpoint difference $p(1-q)$; the selection/retry interaction $v(\{s,r\})-v(\{s\})-v(\{r\})+v(\varnothing)$ is $(1-p)(1-q)$. At $p=q=0.5$, these are shares $[-0.125, 0.375, 0]$, endpoint 0.25 and interaction 0.25. With idempotency, any selected refund creates one effect, so consequence shares become $[-(1-p), 0, 0]$, as for initial action selection; the interaction is zero. These are population references derived from the declared mechanism, not fitted to sampled outcomes. A call cap can refuse the finite estimate despite a defined ideal reference. Unknown evidence or strict unreachable-held decisions do not acquire numeric references from these forms.

A standalone standard-library check uses exact rational arithmetic on a five-by-five (p,q) grid, two idempotency settings and two targets: 100 control cases. It checks all coalition values and six permutation orders against these forms without importing the framework or reading sampled study data. This checks the stated mathematical law, not simulator execution or sampling validity.

TABLE I

Original protective-stop consequence position: nominal 95% percentile-bootstrap contrastive and normal paired-Shapley coverage. Each entry has 128 available, reference-eligible trials.

Retry q	Contrastive covered	Shapley covered
0.1	100/128	107/128
0.5	122/128	121/128
0.9	103/128	103/128

TABLE II

Fresh protective-stop shared-draw coverage: 32 trials per row. Bootstrap/exact-binomial intervals cover the contrastive effect; normal/Hoeffding intervals cover the paired-Shapley coefficient.

q	Bootstrap	Exact binomial	Normal	Hoeffding
0.1	26/32	31/32	23/32	32/32
0.5	31/32	32/32	32/32	32/32
0.9	31/32	32/32	29/32	32/32

V. Audited Evaluation

A. Original calibration and finite analysis

The *protective-stop* fixture forces the factual timeout decision to stop after one committed refund. Fresh continuations retry with probability q, so resampling can introduce a duplicate. The *retry-pivot* fixture instead forces a factual retry; its timeout position is the latest population rescue position when $0 < q < 1$. These are planted scripted decisions, not model-derived causal labels.

The original development study retained 1,280 trials across ten specified mechanisms, with 128 repetitions. Of 223,232 planned branches, 198,690 completed, 4,062 were invalid, 256 failed at the deliberate cap and 20,224 were unstarted. Standalone audit scripts checked raw evidence, seeded outcomes, statistics and accounting. Table I reports a preselected protective-stop consequence position with 128 available, reference-eligible estimates in each row.

Separate finite-distribution calculations yielded protective-stop normal pair coverage of 83.7775% at $q = 0.1, 0.9$ and 92.6377% at $q = 0.5$ for 16 pairs. For interior Bernoulli q, homogeneous draws force an empirical percentile bootstrap interval to a point, giving coverage ceiling $1 - q^K - (1-q)^K$. At $K = 16$ and $q = 0.1, 0.9$, this is about 81.47%. These findings concern the specified adaptation; they do not establish coverage of upstream CAR implementations.

B. Shared-draw conservative comparison

A separately frozen follow-up retained 384 trials across twelve cases, 66,048 planned branches and 87,054 actual scripted calls. Variants reused the same points, draws, seeds and refusals. Exact probability weights over sample counts 8,16,32,64 and eleven probabilities yielded minimum grid coverage 7.72% for the limiting empirical percentile bootstrap, 85.15% for Wilson and 96.68% for Clopper–Pearson. The bootstrap grid is its infinite-resample limit, distinct from the replay's finite seeded bootstrap.

Table II is the protective-stop timeout position, not every case. The full summary includes exact-binomial coverage 30/32 for a retry-pivot case. Finite observed coverage can fall below nominal despite a conservative population method. At $q = 0.1$, mean normal versus Hoeffding width was about 0.180 versus 1.358; at $q = 0.5$, 0.344 versus 1.184. Hoeffding detected the negative reference sign in 0/32 trials at both weaker settings, and 32/32 at $q = 0.9$. Wider intervals therefore impose a substantial detection cost.

The standalone saved-evidence audit checked its recorded derived-artifact inventory and rejected twenty targeted corruptions. A separate SciPy numerical audit covered 1,290 binomial cases and 1,290 endpoint differences, with maximum binomial endpoint disagreement approximately 2.66×10^{-14}. SciPy was an audit dependency, not a framework runtime dependency. Finite grids and synthetic replays do not establish live-model sampling independence or simultaneous attribution coverage.

C. Scripted-simulator interactions and variable length

An independent scripted-simulator enumeration covered four deterministic fresh-choice components, eight held masks, two games and eight cases, retaining 512 planned branches and 718 actual scripted calls. Outcomes came from ordinary payment readback rather than injected payoff values. With selection and timeout retry probabilities $p = q = 0.5$, unsafe retry and exactly-one verification, on-reach consequence shares were $[-0.125, 0.375, 0]$, with selection/retry interaction 0.25. Action-selection shares were $[-0.5, 0, 0]$. Idempotency changed consequence shares to $[-0.5, 0, 0]$ and removed the interaction. This is a declared programmatic mixture, not observed model interaction recovery.

The strict population game remained undefined at interior probabilities; missing provenance, lag and small budgets retained their distinct limits. The independent audit rejected twenty-nine semantic corruptions. The original export manifest and data remain preserved alongside the derived canonical inventory. Two sampler integration controls exercised 32 pairs and 32 endpoint draws, with eighteen coefficients within a broad 0.3 regression tolerance. That tolerance alone is not a calibration result.

D. Repeated scripted-simulator sampling

A separately frozen protocol exercised the actual permutation sampler across all eight declared scripted-simulator mechanisms, both games and 128 fresh repetitions per mechanism/game: 2,048 trials. Each trial planted the same three factual decisions and used 32 forward/reverse permutation pairs, one draw per coalition evaluation and 32 independent draws per endpoint, yielding 320 planned branches. The planted factual path was specified rather than selected by seed search. The cap was 1,000 actual calls including three factual calls, except for the four-call budget control. Collection used normal intervals; Hoeffding and exact-binomial endpoint differences were rescored from identical draws without additional calls, altered points or redrawn refusals. The installed checkpoint, precollection protocol, all compressed trial/branch rows and replicate-zero full evidence with separately stored evaluator truth are retained.

The run retained all 655,360 branches: 536,150 completed, 37,423 invalid, 256 failed and 81,531 unstarted. It consumed 628,535 scripted calls, including 6,144 factual calls, and zero language-model calls. On-reach execution recorded 37,392 branches with inactive requested holds; these are normal-termination records, distinct from strict invalidity and budget exhaustion. Figure 3 retains both games and every branch status, displaying inactive holds separately from the mutually exclusive status partition.

A standalone auditor importing no framework implementation checked the retained trials, branches, 4,210 complete traces and all summary rows. Its receipt binds the frozen source revision and original 52-file manifest. Derived tables retain every row separately from that sealed collection; auditing made no generation calls.

Actual-call cost also depends on availability. The on-reach unsafe $(0.5, 0.5)$ stratum used 42,526 scripted calls for 128 available global-consequence coefficient vectors, or 332.23 calls per vector, including factual capture and independent endpoints. Each vector contains three jointly estimated players. The lagged stratum spent 42,424 calls for zero such vectors; its cost-per-usable vector ratio is undefined. Shared targets and interval variants do not multiply the number of independent usable vectors. These are scripted-call costs, not production latency or financial charges.

Table III reports every declared mechanism for one specified target and position, including unavailable controls. Figure 4 places coverage beside interval width and detection to expose their tradeoff without pooling targets or cases. At $(p, q) = (0.5, 0.5)$, the shared retry estimate had bias -0.00269 and RMSE 0.0611. Normal coverage 118/128 has a 95% Wilson interval $[0.862, 0.957]$; 128/128 has $[0.971, 1]$, not proof of perfect population coverage. The conservative intervals sharply reduced weak-effect sign detection. For the consequence stop dummy, normal false nonzero flags ranged from 0 to 14/128 across the six assessable mechanisms; Hoeffding flags were 0/128 in each. The deterministic zero-reference case yielded 128 normal point intervals; their correctness in that case does not validate point intervals for stochastic outcomes. With 128 repetitions, $[0.862, 0.957]$ also shows why an observed 92.2% rate does not resolve a small departure from 95% coverage. These runs are diagnostic integration evidence, not a precise empirical ranking of near-nominal methods. The finite-distribution calculation supplies a separate analytic check only for its declared independent Bernoulli law.

Endpoint behavior was separately assessed. For the on-reach unsafe $(0.2, 0.8)$ global-consequence endpoint, normal coverage was 92/128 (Wilson $[0.635, 0.789]$), with 35 point intervals and mean width 0.114. Exact-marginal differences covered 128/128, with mean width 0.319. These finite results demonstrate the declared precision tradeoff; they do not establish simultaneous coverage or calibration under live-model dependence.

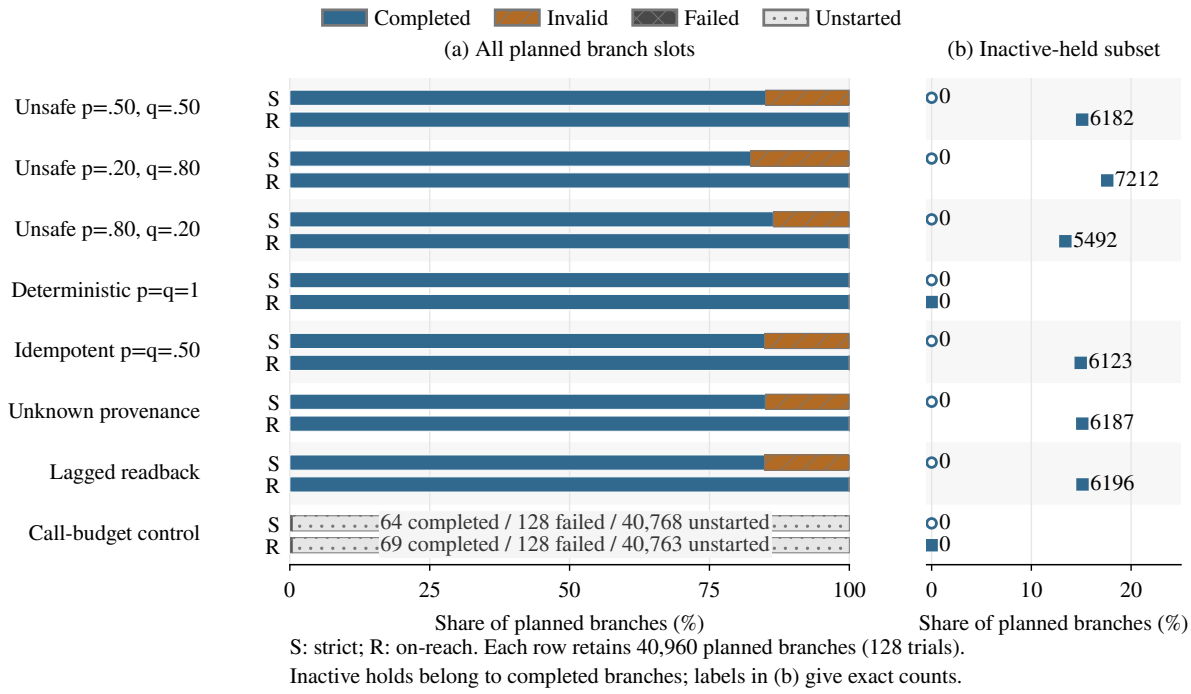


Fig. 3. Disposition of all planned branches across eight mechanisms and two games: 40,960 slots per row. Completed execution includes unknown target evidence and known violations; it does not establish an available coefficient. Failed and unstarted budget-control slots remain visible through exact counts. Inactive holds are a subset of completed branches, shown separately.

TABLE III

ALL EIGHT DECLARED CASES: ON-REACH RETRY COEFFICIENT FOR GLOBAL CONSEQUENCE, 128 PLANNED REPETITIONS PER ROW. N/H DENOTES NORMAL/HOEFFDING. AVAILABILITY A AND POPULATION-REFERENCE ELIGIBILITY R EACH HAVE DENOMINATOR 128. FLAGS DENOTE CORRECT DIRECTIONAL DETECTION FOR NONZERO REFERENCES AND FALSE NONZERO FLAGS FOR ZERO REFERENCES. DASHES DENOTE UNDEFINED QUANTITIES; 0/0 DENOTES AN UNDEFINED COVERAGE RATE. HOEFFDING INTERVALS ARE CLIPPED TO $[-1, 1]$, EXPLAINING THE 0.773 MEAN WIDTH AT $(0.8, 0.2)$ VERSUS THE UNCLIPPED 0.960 WIDTH AT THE TWO OTHER STOCHASTIC UNSAFE SETTINGS.

Case (p, q) or control	Reference	A ; R	Covered N / H	Mean width N / H	Flags N / H
Unsafe $(0.5, 0.5)$	0.375	128 ; 128	118/128 ; 128/128	0.226 / 0.960	128/128 ; 8/128
Unsafe $(0.2, 0.8)$	0.120	128 ; 128	124/128 ; 128/128	0.154 / 0.960	123/128 ; 0/128
Unsafe $(0.8, 0.2)$	0.720	128 ; 128	123/128 ; 128/128	0.218 / 0.773	128/128 ; 128/128
Deterministic $(1, 1)$	0	128 ; 128	128/128 ; 128/128	0 / 0.960	0/128 ; 0/128
Idempotent $(0.5, 0.5)$	0	128 ; 128	123/128 ; 128/128	0.244 / 0.960	5/128 ; 0/128
No provenance $(0.5, 0.5)$	0.375	128 ; 128	120/128 ; 128/128	0.228 / 0.960	128/128 ; 7/128
Lagged readback $(0.5, 0.5)$	—	0 ; 0	0/0 ; 0/0	—	—
Four-call cap $(0.5, 0.5)$	0.375	0 ; 128	0/0 ; 0/0	—	—

In the on-reach game, every lagged-consequence coefficient was refused as unknown; every budget coefficient was refused after unstarted slots, despite the latter's defined ideal reference. Removing provenance preserved global-consequence availability but refused all 128 linked-consequence coefficient vectors. All nondeterministic strict coefficient vectors were unavailable and their population games undefined; the fully deterministic strict control remained defined. One finite strict interaction point at $(0.8, 0.2)$ did not make that population interaction defined. One strict linked-consequence endpoint without provenance was numeric but reference-ineligible, giving coverage 0/0 rather than an accuracy result. Pooled-coalition interaction diagnostics have no confidence interval and supply no coverage or sign-detection result. The full summary retains separate availability, reference eligibility, bias/RMSE, coverage uncertainty, width, flags, endpoints, interactions and refusal rows for every target, position, game and interval variant. Reused targets and variants are not independent replications.

E. Exploratory local-model validation

A frozen installed Qwen3 4B checkpoint completed six validation cases, two repetitions and two unchanged prompt profiles, using 59 actual calls within a 72-call cap. No episode failed or remained unstarted. Model identity, prompt/payload hashes, public traces and the exact source checkpoint are retained. The original and revised profiles induced different action sequences; this is not a pure reporting experiment or an attribution-method comparison.

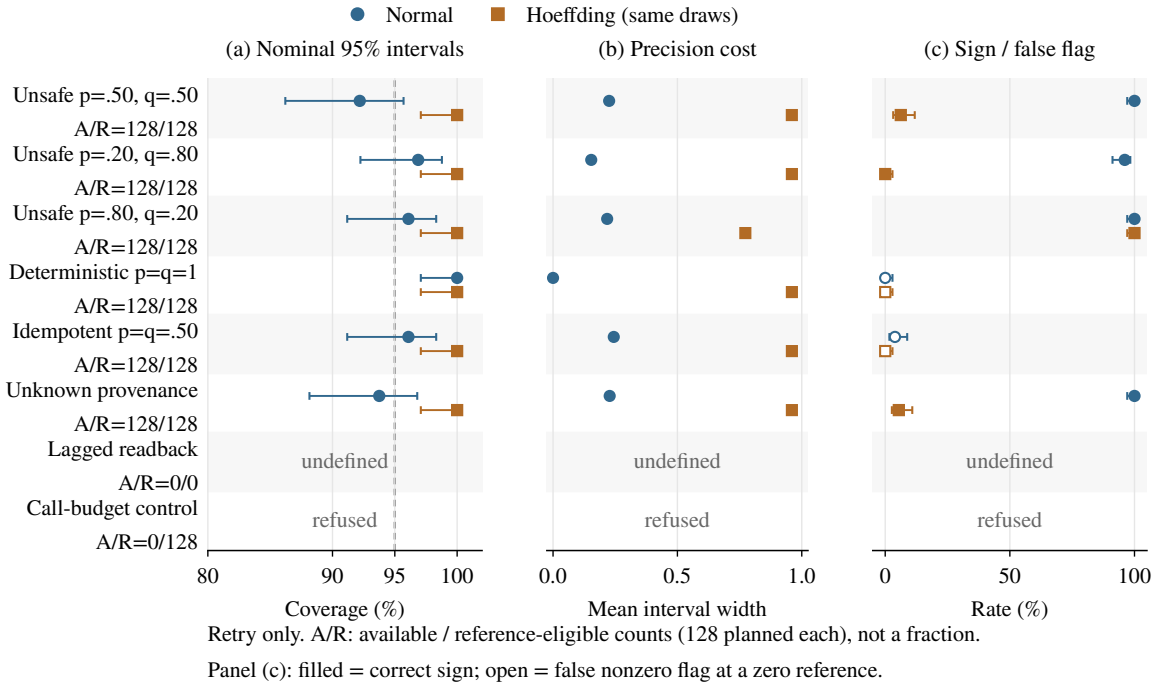


Fig. 4. On-reach global-consequence retry coefficient for all eight cases, using identical draws for both interval methods. A/R denotes available and population-reference-eligible counts, each from 128 planned trials. Coverage and flag whiskers are 95% Wilson uncertainty for repeated-trial rates, not coefficient intervals. Filled flag markers indicate correct directional detection; open markers indicate false nonzero flags for zero-reference controls. Undefined evidence and budget refusals have no zero-valued markers.

TABLE IV
LOCAL-MODEL VALIDATION WITH EXPLICIT CLAIM DENOMINATORS.

Quantity	Original v1	Revised v2
Completed/planned episodes	12/12	12/12
Actual calls/profile cap	36/36	23/36
Definite final status claims	12/12	0/12
Contradictions/assessable claims	6/10	0/0
State satisfied/violated/unknown	6/4/2	7/3/2
Linked satisfied/false/unknown	6/4/2	5/5/2

The revised profile's conditional contradiction rate is undefined, not zero error. The original profile's 6/10 contradiction proportion has a descriptive 95% Wilson interval of approximately $[0.313, 0.832]$. Shared cases, fixed profile order and two repetitions per case preclude interpreting this interval as calibrated uncertainty over a broader population of tasks or models. In the other-actor case, revised-profile readback satisfied state while origin was external and action linkage false. In the expired-request case, the original profile correctly described ineligibility but issued a refund; its failure assessment contradicted achieved effect. Effect satisfaction and eligibility-policy correctness are distinct targets.

A separate unblinded annotation pass within the AI-assisted development workflow marked affirmative processing clauses in 10/12 original and 7/12 revised final rationales. Five of those seven revised rationales also hedged independent verification. A statement that a second attempt committed can be literally true while two effects violate an exactly-one task. These annotations are descriptive; no independent human verification or inter-rater reliability has been established. Unknown structured fields do not ensure prose abstention. The saved auditor rejected seven corrupted smoke copies without additional generation. Fixed profile order, tiny sample size, shared fault conditions and an exposed validation family limit interpretation. The held-out reserve was not generated.

F. Matched three-model factual auditing

A separate internally timestamped protocol froze source `bcd733d8`, model digests, unchanged profiles and wording before generation. Three small local checkpoints (Qwen3 4B, Llama 3.2 3B and Gemma 3 1B), two profiles, two new wording families, seven exposed mechanisms and six repetitions gave 504 planned slots. World/phase seeds were matched across model/profile cells; repetition blocks randomized model, profile and case order. Temperature was 0.2, output cap 384 tokens, with thinking disabled and at most two tool attempts per episode. CPU-only local Ollama used no paid API or real payment. These are fourteen fixed case/wording cells, not 504 independent task samples. Family, size and quantization are confounded; the earlier held-out reserve remains unused. The frozen six-hour/1,512-call limits were not exhausted.

Exact checkpoints: qwen3:4b-instruct-2507-q4_K_M; llama3.2:3b-instruct-q4_K_M; gemma3:1b-it-q4_K_M.

Five offline auditors score identical completed episodes, with no extra model call: last public response; final structured assessment; readback without provenance; an independently implemented contract/provenance reference; and Consequa-X. The last two share information. Ledger gold uses a separate discrete-time predicate; the reference uses an event sweep. This compares factual information budgets without extra model calls; the separately frozen trace-judge comparison follows below. This five-auditor analysis does not measure causal attribution or a verify-before-retry controller.

All 504 slots started; 503 completed and one Llama v2 missing-provenance slot failed strict-JSON parsing. That failure was retained without replacement. There were 896/1,512 attempted calls, all with received responses, and no transport failures, interruptions or unstarted slots. Collection finished normally in approximately four hours 58 minutes. Definite coverage divides claims by completed gold-assessable episodes; claim error divides wrong by definite claims; correct yield divides correct by assessable episodes. Full planned operational yield retains all 504 slots. Zero denominators remain undefined and abstentions do not count as correct.

Figure 5 retains every auditor. Response-only made 201 state claims, with 53/201 wrong and 148/503 correct yield; structured self-report made 252, with 148/252 wrong and 104/503 yield. Readback, reference and Consequa-X each made 431/503 correct state claims, with 0/431 observed errors and 72 lag abstentions. For action linkage, response-only had 53/201 wrong, self-report 136/252, and readback-only abstained on all 503. Reference and Consequa-X made 385/503 correct linked claims with 0/385 errors; the remaining 118 comprise 72 lag and 46 omitted-provenance abstentions. Planned state/linked yields for Consequa-X were 431/504 and 385/504.

Origin gold was assessable for 310/503 completed episodes, including five excess-effect episodes that failed the exactly-one state contract. Reference and Consequa-X made 216/310 correct definite origin claims (144 confirmed, 72 other-actor), with 0/216 wrong and planned yield 216/504. The 94 assessable origin abstentions comprise 48 lag and 46 omitted-provenance cases; the other 193 completed episodes had unknown ledger-origin gold. Other auditors made no origin claims. Complete reference/Consequa-X predictions, including abstention and origin, agreed on 503/503 episodes. This parity supports implementation correctness, without demonstrating advantage over an equally informed method.

Execution diagnostics were added descriptively after the first-six no-action audit, without changing generation, primary metrics or stopping rules. Only 334/503 completed episodes dispatched an agent action (334/504 planned), with 392 tool attempts and 58 retries. Gemma dispatched 0/168 completed episodes; Llama dispatched 166/167, with zero retries; Qwen dispatched 168/168, with 47 v1 and 11 v2 retries. An external actor can still create an effect during a no-action episode. Empty-ledger correctness must not be described as refund execution or fault handling. Figure 6 places these execution rates beside semantic yield. Raw byte journals, every disposition, source-bound inventories and condition/model/profile fractions are retained. No pooled significance, population confidence interval, model ranking or independent human annotation is established by this fixed simulator collection.

G. Same-evidence local trace judge

A separate protocol froze source `be842879`, aliased inputs, prompt, checkpoint and budget before judge generation. It selected the 84 original schedule slots with repetition zero after factual collection, retaining every selected input; these span fourteen fixed wording/mechanism cells and seventeen distinct public evidence patterns. The judge was the installed Qwen3 4B checkpoint used in the factual study, at temperature 0.2, a 192-token cap and thinking disabled. Each input received one call without repair or replacement. The allowlisted input contained action/contract, dispatch/response context and captured readback with scope, completeness, watermark and provenance. Equality-preserving aliases hid identifiers; generator/fault labels, reported assessments, rationales, verification/explanation outputs and evaluator truth were excluded. The strict output schema requested only state, action-linked satisfaction and origin.

All 84 selected traces completed in 84 local calls, with no failed, interrupted, unavailable or unstarted slots. The judge made 60/84 definite state claims: 9/60 were wrong, all false positives, giving 51/84 correct yield and 24 abstentions. An automated public-reference rule qualified 43 state claims as evidence-supported; eight other correct guesses lacked that support. All 84 action-linked outputs were unknown: claim error is undefined, not zero. Of 52 origin claims with assessable gold, 40 were correct and 12 wrong; eight additional definite origin outputs had unassessable gold and receive no invented true/false label. Of the 40 correct origin claims, 24 were supported by the public rule and 16 were unsupported correct guesses.

Reference and Consequa-X remained exactly equal, making 72/84 correct state, 64/84 correct action-linked and 36/52 correct assessable-origin claims, with no observed wrong claims; selected-slot origin yield was 36/84. Hindsight correctness therefore remains distinct from source-qualified explanation. The subset is not held out or externally preregistered; repeated evidence, one small judge sharing a generator family, and rule-based qualification limit inference. This comparison supplies neither independent human annotation nor population-level evidence about LLM judges (Fig. 7).

H. Maintained benchmark tool-state transfer

A separately frozen, model-free experiment used the MIT-licensed maintained tau3 corpus at `96a3ef92`: all 114 retail and 50 airline `base` tasks in source order, with unchanged native tools, environments and DB/environment evaluator [20], [21]. Three development tasks (retail 0; airline 19 and 20) exercised the adapter; none were removed from the complete corpus. Four fixed controls replayed reference actions, omitted all writes, omitted writes with a fabricated success acknowledgement, or altered the first write's identity/scope

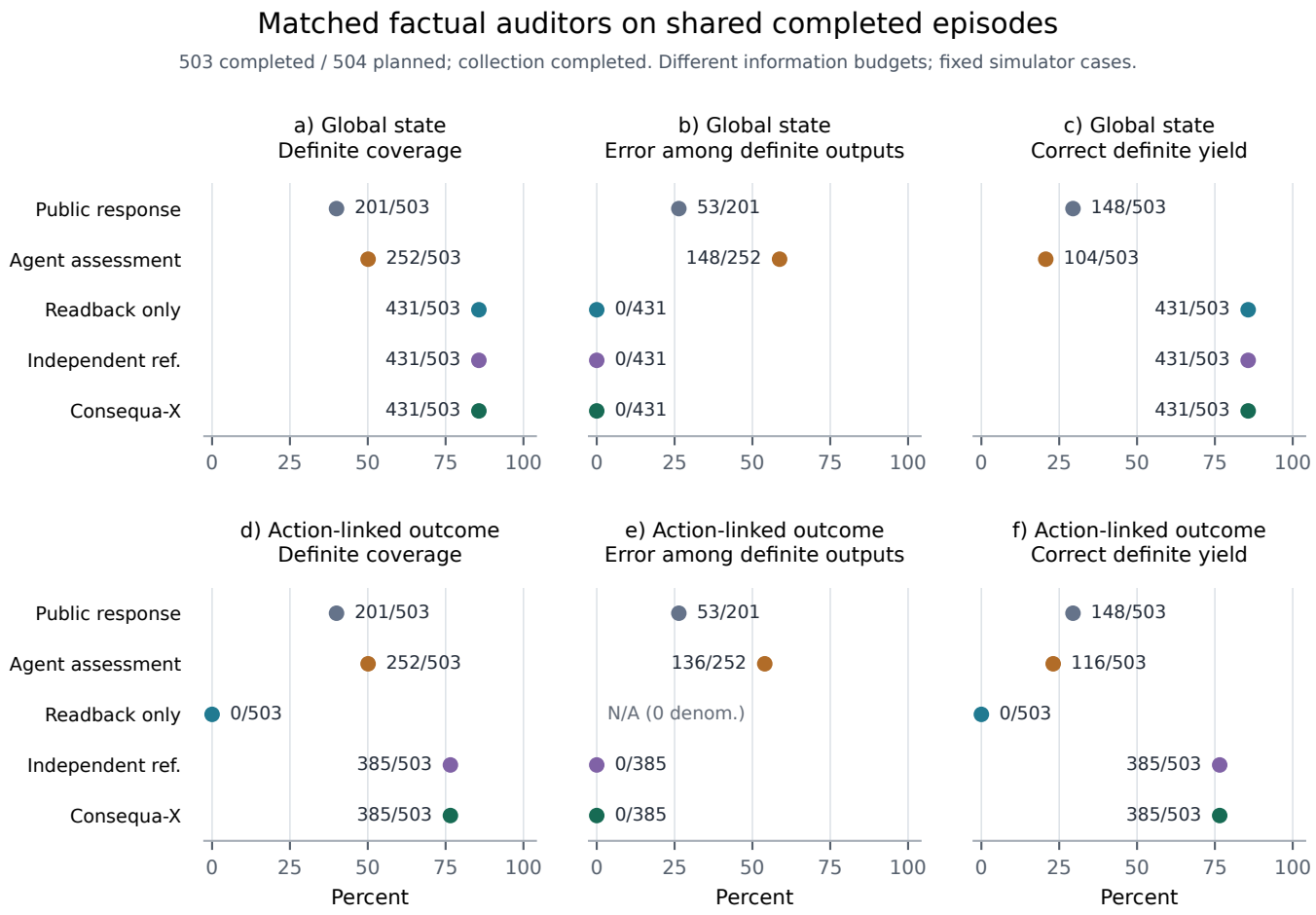


Fig. 5. Matched factual auditors on shared completed episodes. Exact fractions separate definite coverage, error among definite outputs and correct yield for global state and action linkage. Auditors have declared different information budgets; reference and Consequa-X share the same information. Points are descriptive, without independent-task confidence intervals.

while retaining later calls. All 656 cells completed, with zero failures or unstarted cells and no model calls. Source/data, dependency bodies, protocols, starts and result hashes were audited against the frozen source `7f6ef0c`.

Application contracts used each intended operation's payload and native public preflight lookup. The read-only auditor received subsequent native order, user or reservation lookups, without the complete reference action list, gold DB or native reward. Contracts checked requested record fields, not full financial settlement, policy or causal origin. The separate native evaluator compared complete task DB end states after replaying the reference on a fresh environment; it does not require a unique action sequence. No dynamic user simulator, autonomous agent, communication criterion or NL judge was evaluated.

In reference/omit/fabricated-acknowledgement/alter-first-write order, native DB matches were (114, 11, 11, 13)/114 for retail and (50, 24, 24, 24)/50 for airline. Corresponding immediate field satisfactions were (174, 2, 2, 72)/176 and (49, 0, 0, 23)/49. Ten retail and 24 airline reference tasks contained no mutating call. All 900 operation captures were valid; every withheld-readback check returned unknown, origin remained unknown and assessment alignment unassessable. Native reference DB equality of 164/164 is not task correctness: retail task 64 has a rejected exchange followed by a successful pending-order modification; task 105 has a balance-rejected exchange and an unmeasured NL criterion requiring both exchanges. These are retained without a dataset-bug claim. In tasks 22/110, omitted writes can satisfy two individual address-field predicates because the original address already matches a requested or restored state, without proving mutation. The experiment supports cross-domain interface transfer, with task-level and operation-level targets kept separate; it does not establish production effectiveness or

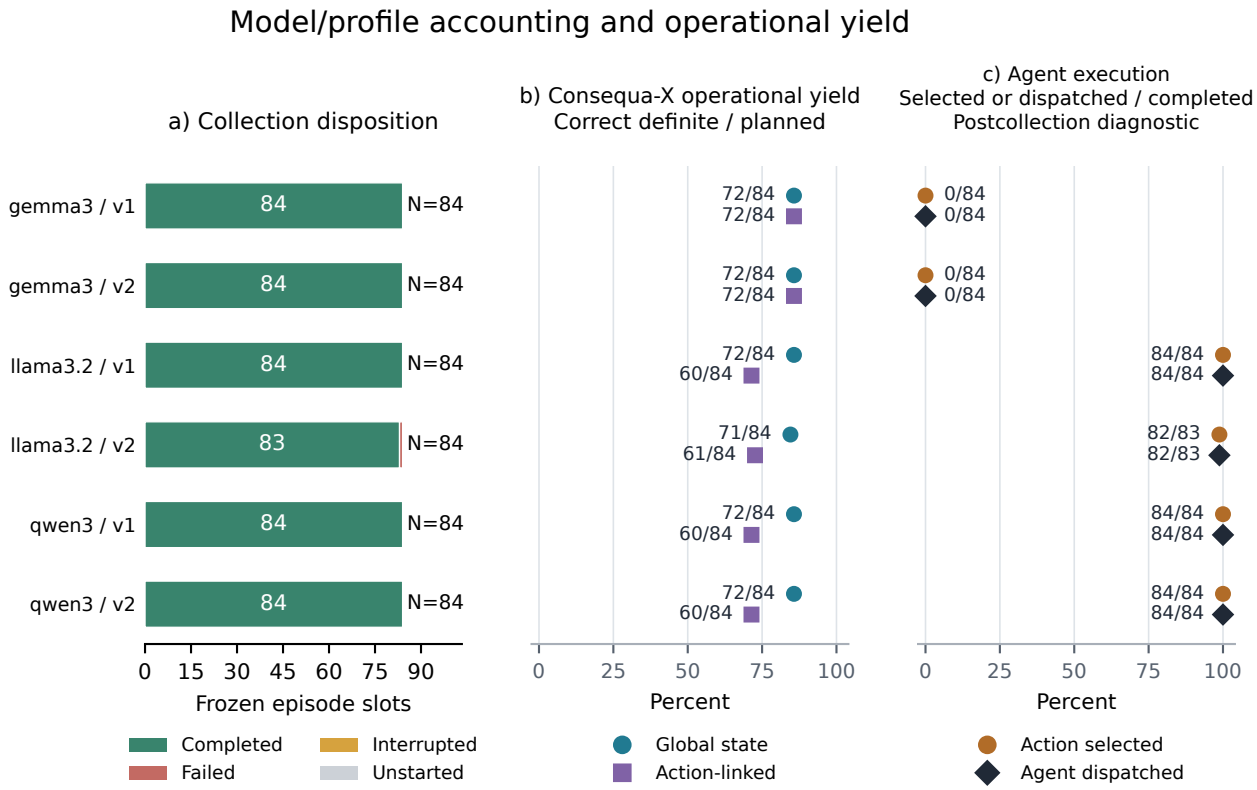


Fig. 6. Model/profile accounting and execution. Operational yields retain all 84 planned slots per group. The dispatch panel uses completed episodes and is a post-freeze descriptive diagnostic. A no-action episode does not exercise a dispatched fault. Full checkpoint tags and all fractions are retained with the source figure inventory.

end-to-end benchmark performance (Fig. 8).

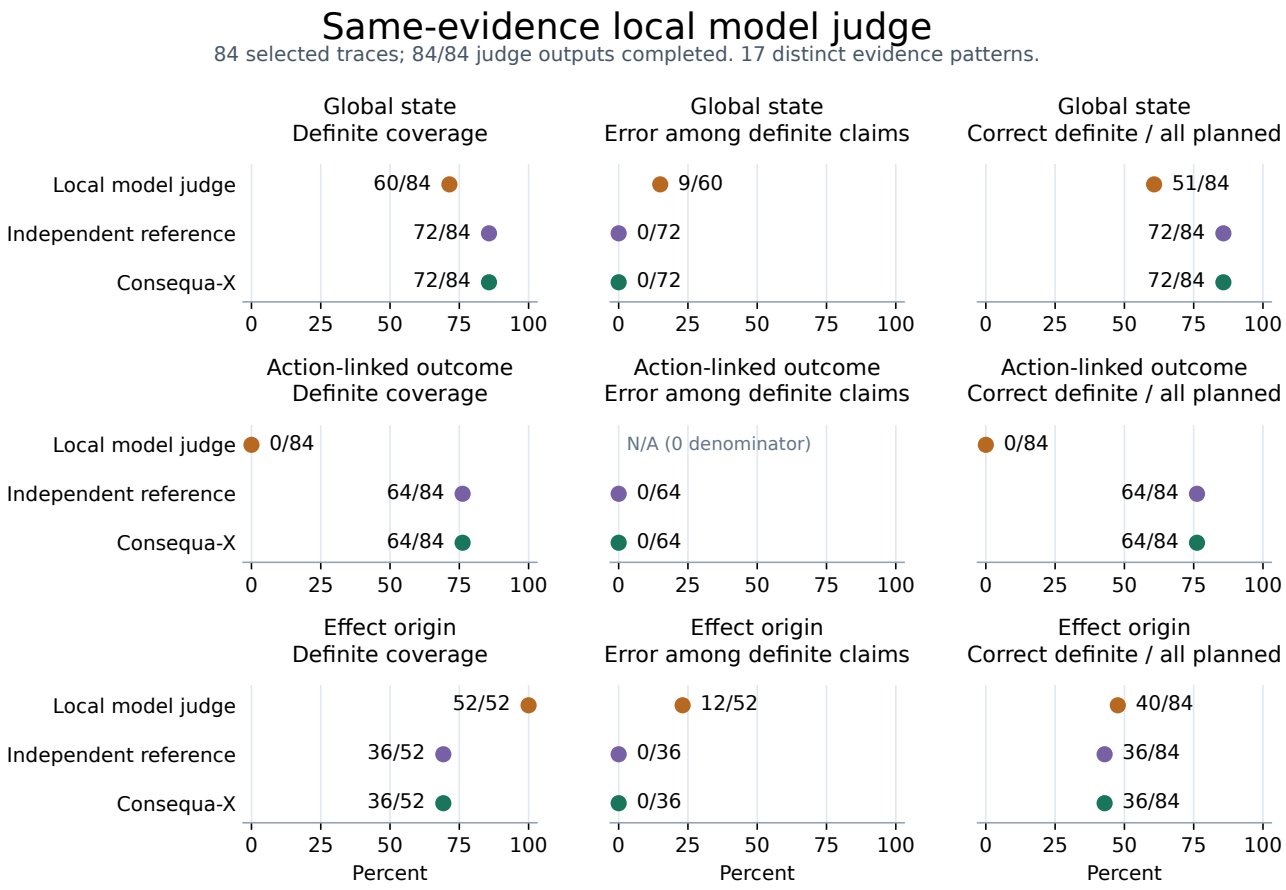
I. Pinned native CAR comparison

The earlier studies compare independent adaptations. A separate prospective protocol at local source `940ebfc` executes native CAR 0.2.0, revision `db90fa28`, without changing its contrastive, permutation, NumPy percentile-bootstrap or SciPy normal-interval algorithms [22]. The bridge supplies the same isolated refund world, planted factual selection/retry/stop, continuation policy, outcome contract, captured readback and on-reach game to both engines. Shared branch/phase seeds and permutation order match draws rather than only nominal budgets. Native CAR postdates the cited paper version; this is compatibility testing of the pinned implementation through the declared bridge, not a reproduction of the complete CAR paper or a live-model attribution benchmark.

Eight fixed mechanisms, two algorithms and 64 repetitions form 1,024 paired groups, each containing two engine trials. Contrastive sampling uses 32 draws per factual position (96 slots per engine trial); Shapley uses 32 antithetic permutation pairs with one rollout per coalition and 16 extra independent endpoints per arm (288 slots). There are 393,216 planned engine draw slots. Independent finite Bernoulli enumeration supplies population references. Repeats characterize fixed simulator settings: nominal 0.95 coverage has Monte Carlo standard error about 0.027 at 64 repeats, without independent-task or multiplicity-adjusted inference.

Both engines share a cap on actual fresh policy calls, counting factual capture and supplemental endpoints; held decisions consume no fresh calls. The four-call control cannot increase its cap. Native rollout-budget charges, deterministic prefix reconstruction, public tool re-execution and wall time are separate cost fields. Native unknown labels and truncated vectors remain raw diagnostics. An identical complete-method gate requires known factual evidence and every planned draw before either primary coefficient vector is released; it does not establish population-game validity from a finite sample. Predeclared budget-start/refusal timing differences remain visible, while other paired semantic mismatches stop collection without redraw.

Collection closed with all 1,024 paired groups and 2,048 engine files retained, including all 393,216 canonical planned draw slots and zero redraws. The independent audit checked every file, paired slot and source binding and recomputed



One frozen Qwen3 4B judge on a fixed repetition-zero subset. Gold-unassessable origin is excluded only from semantic coverage/error denominators. Failures and abstentions remain in all-planned yield. The judge additionally made 8 origin claims with unknown gold; these are not scored as correct or wrong. N/A is undefined, not zero error. Descriptive fractions; no independent-task confidence intervals.

Fig. 7. Same-evidence local trace judging. Coverage uses gold-assessable episodes, error uses definite claims with assessable gold, and correct operational yield retains all 84 selected slots. Undefined zero-claim error remains undefined. Eight extra origin claims with unassessable gold are retained separately. Points compare one local judge and two equally informed deterministic implementations, without independent-task intervals or a general model-superiority claim.

all summary rows without importing either implementation. All factual traces and successful fresh-call totals match in all 1,024 pairs. Both engines release 640 paired coefficient vectors and withhold 384; every eligible paired point estimate agrees exactly. All non-budget draws agree semantically. Unknown-provenance, lag and budget groups remain visible with zero primary availability.

At the $p = q = 0.5$ timeout-retry coordinate, native bootstrap coverage is 59/64 with mean width 0.331, compared with 60/64 and 0.329 for the independent bootstrap adaptation. Its Clopper–Pearson alternative covers 62/64 at width 0.356. Native and adapted normal Shapley intervals both cover 60/64 at width 0.225; Hoeffding covers 64/64 at width 0.960. Finite bootstrap resampling implementations remain distinct. Figure 9 displays all eight mechanisms at that coordinate; all three coordinates and 144 summary rows remain in the artifact. These descriptive repeats do not establish a method ranking or precise population calibration.

The audit found one collector deviation: in 64 native contrastive budget groups, concurrent zero-spend refusals coalesce under one failed canonical slot. Their 6,144 canonical slots are 40 known, 64 failed and 6,040 unstarted. These labels do not identify actual coroutine starts one-to-one: source and retained refusal records support 2,048 first-coalition invocations, with 40 completed and 2,008 refused. Individual refused-coroutine identities were not retained. All 64 affected vectors are already ineligible; no accepted estimate or successful-call cap changes. This limits refused-attempt attribution, without changing the frozen records or silently amending the plan.

Each engine uses 209,666 successful fresh scripted policy calls, including factual capture and endpoints, with zero live model calls. The native bridge records 845,672 determinis-

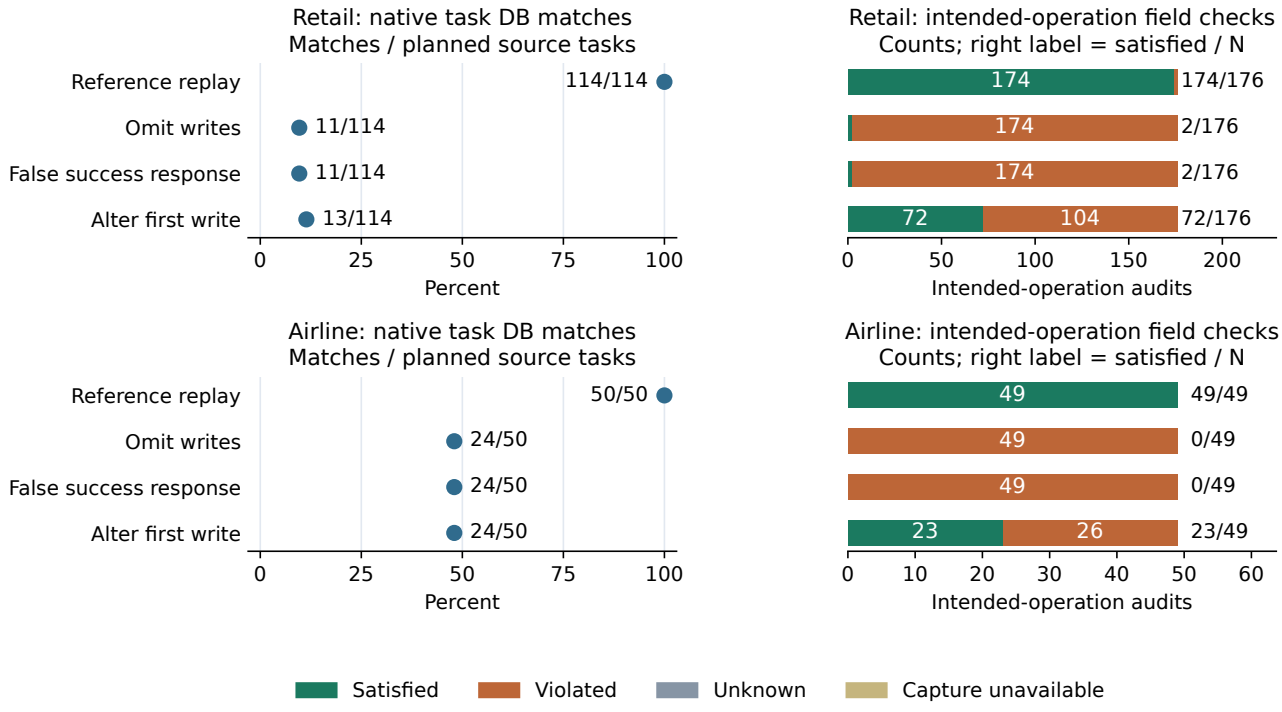


Fig. 8. Model-free benchmark tool-state transfer. Task-level native DB matching has denominators 114 retail and 50 airline tasks per control; immediate requested-field checks have denominators 176 and 49 operations. These targets and units differ. Reference DB equality is self-replay consistency; field satisfaction is not independent predicate accuracy. This figure reports no autonomous agent benchmark score or pass@k result.

tic reconstruction/probe worlds and 968,304 tool dispatches. These counters are adapter work, separately recorded rather than independently reconstructed from full bridge histories. CX completed replay worlds and preflights have different scopes. Wall times include worker contention and bridge overhead, so they do not establish intrinsic speed.

J. Offline conformance across the breaking 2.0 release

The preceding experiments remain frozen on 1.1.0/schema 0.1. To assess a narrow migration boundary, source `f64bf49` and a separate protocol were committed before processing all 503 retained factual traces. The original 504-slot schedule and its failed strict-JSON policy output remain unchanged; no replacement trace was generated. This is an exposed-data, post-hoc conformance study, with zero new model calls or action executions. Isolated 1.1.0 and 2.0.0 environments load the exact pinned implementations. Each legacy observation is bound by its recorded ID, source and watermark; fresh verification uses that same stored observation rather than a new readback. Each version's action-linked target helper is used, without substituting a nonempty target-ID list for its provenance-qualified result.

All 503 traces migrate, round-trip through schema 1.0 and preserve the declared legacy structural projection. Migration preserves recorded conclusions by construction; separately, fresh 2.0 verification agrees with fresh frozen-1.1.0 verification on all legacy receipt fields, state, origin, action-linked target and assessment alignment. The recorded old analysis and explanation also agree with their frozen recomputation. Every primary comparison is 503/503 with zero processing failures or unavailable comparisons. This is 503 usable inputs per 504 historical slots, not a disagreement assigned to the missing input. Agreement includes unknown: fresh state is satisfied on 257, violated on 174 and unknown on 72. Alignment is consistent on 48, contradicted on 108 and unassessable on 347.

Diagnosis codes are deliberately not a canonical invariant. They agree on 228/503 traces and change on 275: 2.0 adds `UNASSESSABLE_ASSESSMENT` in all 275 and removes `CONSISTENT_OUTCOME` in 24 of those. Migrated and freshly verified explanations show the same changes. Target-match metadata is nonempty on 144 receipts, empty on 287 and unavailable on 72; these categories are neither action-linked

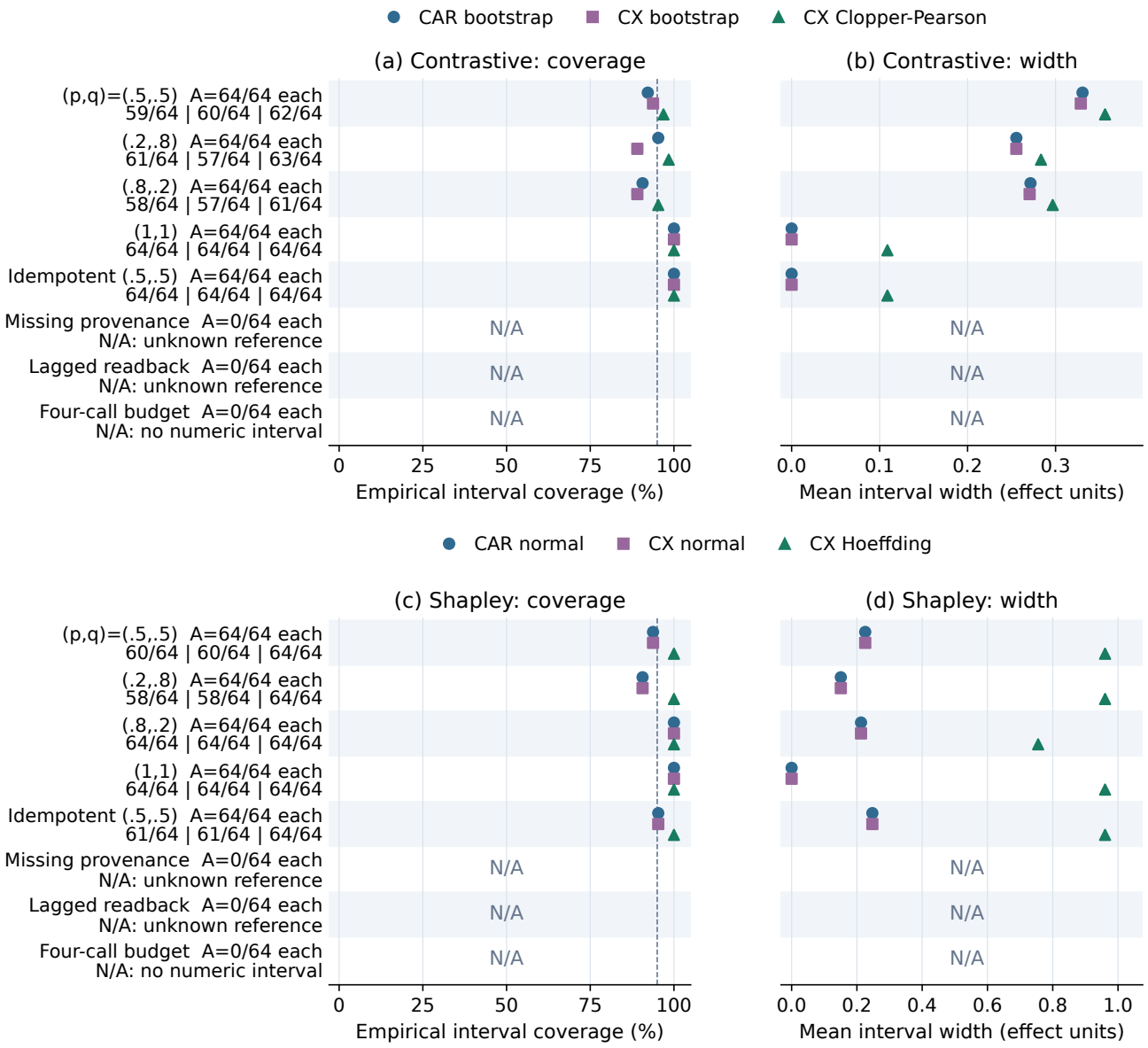


Fig. 9. Pinned native CAR and frozen Consequa-X 1.1.0 on the same timeout-retry coefficient and replay draws. Every fraction uses 64 repetitions within a fixed mechanism; A reports primary vector availability. Numerical coverage uses reference-eligible available intervals, while missing evidence and budget refusal remain undefined. Fraction order is CAR / Consequa-X original / Consequa-X conservative. The budget control has coalesced native refusal records, disclosed separately; none yields an eligible estimate. These are conditional implementation/calibration checks, without a method ranking or task-population claim.

accuracy nor independent labels for the new fields. All 503 migrate to `relative_ms`, because schema 0.1 did not record a clock origin. The receipt-reproduction gate checks status, origin, effect IDs and attributable IDs; it does not certify every added metadata field.

An independent artifact auditor checks hashes, all slot

identities, structural projections and refund evidence arithmetic without importing the framework or collection runner. Its JSON-level checks are distinct from the runner's recorded framework round-trip assertion. This single-action refund corpus supplies no conformance claim for general multiple-action traces, custom verifiers, polling, controllers or the changed replay transport. Canonical agreement is a migration result, not new predicate accuracy or end-to-end equivalence of the two releases.

VI. Discussion and Limitations

The evidence protocol supports different, inspectable claims about responses, state, origin and intervention effects. Its refusal rules withhold affected numerical outputs when captured factual evidence or a planned draw is invalid or unavailable, but can sharply reduce availability. Availability can depend on interventions and must remain visible beside numerical results. Conservative intervals do not make undefined games numeric, resolve missing evidence, or remove the need to assess the actual sampling law.

The refund experiments remain a narrow scripted simulator, and no real payment API was evaluated. The local trace judge adds a same-evidence model comparator; the native retail/airline replays add maintained benchmark tool-state transfer, not autonomous benchmark performance or a second production-domain study. Declared source authority and watermarks are not authenticated transactional guarantees. The three small generator checkpoints and one overlapping judge cover exposed development/validation mechanisms. Fresh wording, seventeen judge evidence patterns and canonical task/control replays do not establish held-out fault transfer or independent task-population inference. State and linked targets sharing outcomes are not independent replications. Claims about hidden reasoning cannot be inferred from reported assessments.

Three evaluation gaps remain. First, extend the completed factual ablations, local trace-judge comparison, model-free tool-state transfer and pinned native CAR bridge to a live verify-before-retry controller, full autonomous benchmark runs and joint causal/structural attribution. The bridge tests native estimators on a matched scripted game; it does not reproduce CAR's full live-agent evaluation. Further comparisons require matched information, targets and call budgets; state satisfaction, source-supported origin and causal intervention effects need distinct reference labels. Report duplicate effects, false success, availability, attribution accuracy and actual cost. Joint context/fault attribution must expose infrastructure variables to every applicable method. The current trace judge is one small local checkpoint, not an independently validated population of stronger judges.

Second, assess bounded polling on an eventually consistent readback: poll until its declared watermark covers the horizon or a predeclared wait budget expires, then retain unknown. Report availability against wait time, read count and latency. Polling cannot promise finality or exactly-once execution under arbitrary late commits. The retained lag control uses one insufficient readback and yields zero consequence estimates; a polling result has not been measured. A safe real-service study would use provider test mode, isolated accounts and injected acknowledgement faults; counterfactual replay would remain in an isolated simulator or a separately validated digital twin.

Third, sample independently authored held-out task and fault-composition families, with an explicit task-population precision analysis and independent blinded double annotation. The 504-slot study supplies conditional repeated observations, not 504 independent task samples. More seeds cannot replace domain diversity. At least 1,000 independent repetitions per selected simulator setting would better resolve interval coverage, with simultaneous uncertainty and modern bounded-mean comparators. No polling effectiveness, full live-agent causal-method comparison or human annotation agreement is claimed here. Offline 2.0 conformance reuses exposed saved evidence; its 503 agreements do not add independent task samples or validate new features.

The retained cost record counts actual scripted calls and refusals; the repeated study used 628,535 calls and zero model calls. It does not measure per-call verification latency, trace-size overhead, real-API charges or cost per usable production estimate. Those measurements belong in the next evaluation.

VII. Conclusion and Artifact Availability

Consequa-X provides a reusable evidence core and auditable replay adapters for separating reported assessments, external outcomes, origin and intervention effects. The retained studies expose uncertainty failure, precision costs, variable-length restrictions and prose/structured-status disagreement. Matched factual auditing additionally measures the limits of response/status proxies and agreement with a same-information reference, including abstentions. The same-evidence local judge exposes unsupported correct guesses, while maintained benchmark replays separate task DB equality from immediate operation fields. Neither supplies a general method ranking or independent human labels. Pinned native CAR runs additionally show exact point agreement for 640 eligible paired vectors, with unavailable groups and a refusal-recording deviation disclosed. Offline 2.0 migration preserves canonical conclusions on 503 retained traces while changing diagnosis codes on 275; the earlier experiments retain their evaluated version and source pins. The useful result is a reproducible protocol/software artifact with measured limits, without claiming a new estimator or general method superiority.

Version 2.1.1, now branded ConseqaX, is publicly available under MIT on PyPI [24], with source and retained research records in the public repository at https://github.com/Devrajsinh-Jhala/consequa-x. Its release record binds the distribution hashes to build source `b73ecd0`. The earlier breaking 2.0 release introduced schema 1.0, multiple actions and assessments, declared time bases and immutable keyword-only records. Target matches and findings move into verifier receipts; explanation selects the latest assessment per action. Migration has the separate offline conformance check above.

Capture, polling and retry-advice interfaces are later product capabilities, not interventions measured by the frozen 1.1.0 experiments. The 2.1 line adds verifier composition, OpenTelemetry interchange and optional framework integrations. Models, the built-in refund verifier, explanation, target helper and numerical estimator modules remain unchanged from the measured 2.0 source, but the public verification wrappers now accept multiple readbacks. The 2.0 result therefore does not validate all 2.1.1 execution paths. Release-specific installed-artifact checks cover Windows Python 3.11–3.15 and one emulated Alpine Linux/Python 3.14 guest; macOS remains untested. The release retains one live engineering smoke run per integration: OpenAI Agents SDK and LangChain use local Ollama, while Claude Agent SDK uses authenticated remote Haiku with an in-process SDK MCP tool server. These runs test wiring, not comparative task accuracy, deployed MCP services or OpenTelemetry collectors. They are separate from the frozen scientific experiments. A DOI-backed immutable research archive has not been established.

The earlier experiments retain source pins `9e2dcbb` (original calibration), `0d8ec0b` (shared draws), `2c4adfe` (enumeration/local validation) and `0bd744d` (repeated sampling). The three-model factual study used `bcd733d8`; the judge protocol declared `be842879`; tool-state transfer froze `7f6ef0c`, with maintained external source `96a3ef92`; native CAR used `940ebfc` with upstream `db90fa28`. Full revisions, protocol hashes and inventories remain study-specific. Those experiments used the 1.1.0 runtime and schema 0.1 [23]. The 2.0 estimator modules retain their numerical implementations, but record models, verifiers, explanation and replay transport have changed. The offline migration study uses separate source `f64bf49` and a separately committed protocol; byte identity of estimator modules alone does not establish end-to-end equivalence. Later releases do not rerun, reseal or relabel the frozen studies. Readback trust and completeness remain application responsibilities. GitHub Actions remains disabled. Framework imports and offline examples require no model credentials or required runtime dependencies. All reported payments are simulated; model weights are identified rather than redistributed.

AI Assistance and Verification Disclosure

Generative AI assistance was used for software development, research-source inspection, rationale annotation and manuscript drafting. Recorded automated checks cover retained-evidence accounting, analytic references, numerical implementations and document compilation. They do not constitute independent human replication or blinded annotation. The rationale labels remain unblinded and exploratory. The author is responsible for the paper's claims and any venue-required disclosure; no independent human validation is asserted.

Appendix A
Declared-Game Properties and Test Evidence

Strict-game partiality. In the declared three-position mechanism, let $N = \{s, r, t\}$ be selection, timeout retry and final stop. If $p < 1$, the coalition $\{r\}$ requests a retry that is unreachable with probability $1 - p > 0$ when fresh selection declines. If $q < 1$, coalition $\{s, t\}$ holds selection and requests a final stop that is unreachable with probability $1-q > 0$ when the fresh timeout decision stops. With compatible decisions, normal execution, adequate budget and known outcomes, the full strict coalition game is structurally defined if and only if $p = q = 1$ in this mechanism. In particular the full strict game is partial at every interior (p, q) and ordinary full-game Shapley is undefined there. Conditioning on valid replay would define a different game. A finite sample with no invalid draw cannot establish zero population invalid probability.

On-reach allocation. Fix finite factual positions, an intervention mapping and a continuation law. Suppose each coalition terminates almost surely, all reached held decisions are compatible, and a bounded measurable numeric outcome exists almost surely. Ignoring holds beyond normal termination then defines $v(S) = \mathbb{E}[X_S]$ for every coalition. Its ordinary Shapley values obey efficiency, symmetry, the null-player property and linearity. To see efficiency, sum the marginal contributions along each permutation: they telescope to $v(N) - v(\varnothing)$, and averaging preserves the equality. Relabelling permutations gives symmetry; zero marginals give the null-player property; linearity follows from linearity of each difference and of averaging. These properties belong to this declared game. They establish neither causal identifiability nor nonnegative normalized responsibility. Unknown observations, incompatible holds and provider failures remain outside these assumptions. Finite independently sampled coalition values can give a nonzero estimate for a true null player, and an independently sampled endpoint need not equal the sum of finite estimated shares. Uniform forward permutations and their reverses each have a uniform marginal. With unbiased coalition samples, their paired average is unbiased for the declared Shapley value. Independent pairs, rather than individual walk steps, are the sampling units for the bounded-mean interval.

Test mapping. C1 is exercised by external-actor and absent-provenance tests in `test_verification.py` and conflicting-observation tests in `test_graph_identity.py`. C2 is exercised by `test_refund_interactions.py`, the exact-rational analytic checker, and retained enumeration controls. C3 is exercised by unknown-draw and exhausted-budget tests in `test_decision_shapley.py`, plus the saved-evidence branch audit. Tests establish behavior in their stated cases, not correctness of an untrusted source or a real-service policy.

Appendix B
Supporting Audit Inventory

Raw assertion counts are retained for reproducibility rather than treated as evidence strength: 3,283,523 for the shared-draw audit, 27,277 for the enumeration audit, 44,698,925 for repeated sampling, and 2,227 for the local validation. These

counts depend on data volume and auditor implementation. More informative checks include independently recomputed summaries, exact-rational analytic references, source-bound inventories, refusal denominators and rejected semantic corruptions. None supplies independent human verification.

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